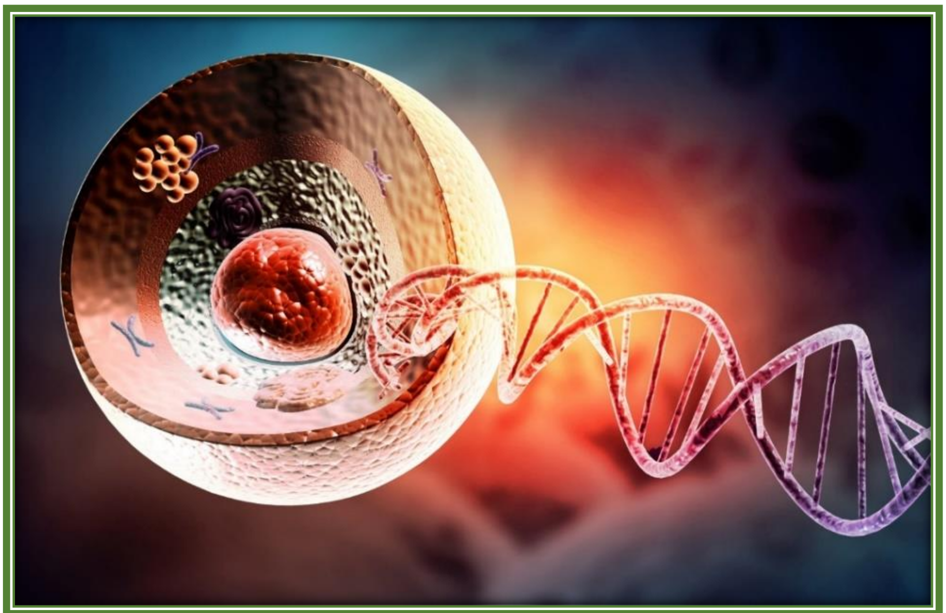
1 The Fundamental Unit of Life



"Cell is the structural and functional unit of living organisms."

Introduction

Cell is the basic structural and functional unit of living organisms. Cell biology is branch of biology that deals with the study of cell in all respect of its structure and function.

Discovery of cell

(i) Robert Hooke (1665) observed dead cell which resembled honeycomb like structures in cork (comes from bark of tree). He called these boxes 'cell'. He published what he observed under his self-designed microscope (primitive microscope) in his book 'Micrographia'.

(ii) Anton Van Leeuwenhoek (1674) was first to discovered free living cell in pond water with improved microscope.

(iii) Robert Brown (1831) discovered the nucleus in cell.

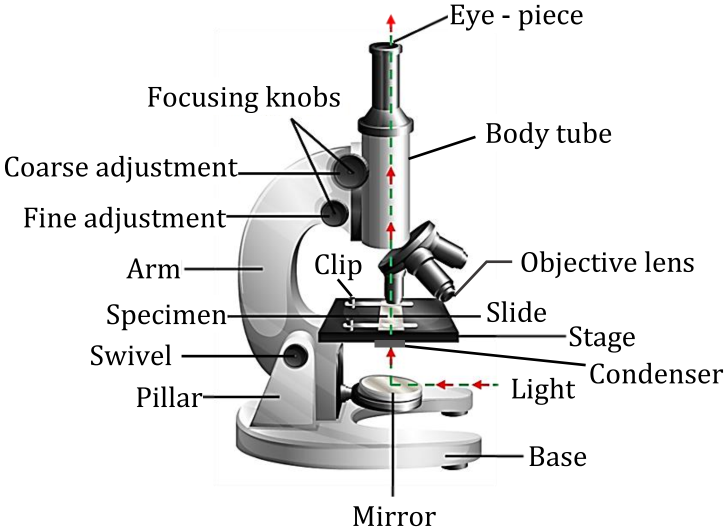
(iv) Dujardin (1835) discovered living fluid substance of cell and named Sarcode.

(v) J. E. Purkinje (1839) coined the term protoplasm for the fluid substances of the cell. It is living matter present inside the cell.

(vi) Huxley (1868) called protoplasm "The physical basis of life".

> The word cell came from latin word 'cellula' meaning a little room. Plant cell was the first cell to be discovered by scientists.
>
> **SPOT LIGHT**

(vii) Rudolf Virchow (1855) – German pathologist established that all cells arise from pre-existing cells ("omnis cellula e cellula").



Compound microscope

- A compound microscope is an optical instrument for forming magnified images of small objects consisting of an object lens and an eye piece, both lenses mounted in the same tube.
- The discovery of electron microscope by Knoll and Ruska, in 1939 made it possible to observe and understand the complex structure of cell and its organelles.

Cell Theory

The "cell theory" was formulated by two biologists, M.J. Schleiden (1838), and T. Schwann (1839). According to them, all the plants and animals are composed of cells and that the cell is the basic unit of life. The cell theory was further expanded by Virchow (1855) by suggesting that all cells arise from pre-existing cells.

Cell theory, states that

(i) All living organisms are composed of cells and products of cells.

(ii) All cells arise from pre-existing cells.

Size of cell: Normal size of human cell - 20 μm to 30 μm in diameter. 1 μm = 1×10^{-6} m

(i) Largest isolated single cell in animals is an Ostrich egg.

(ii) Longest cell in animals is Nerve cell.

(iii) Smallest cell is PPLO-Pleuro Pneumonia Like Organism (Mycoplasma, 0.3 μm in length).

- Bacteria measures 3-5 μm, while viruses are still smaller and do not have a cellular structure.
- The longest living cell in the human body are nerve cells (or neuron) which can live for entire life time.
- Human red blood cells - 7 μm in diameter.

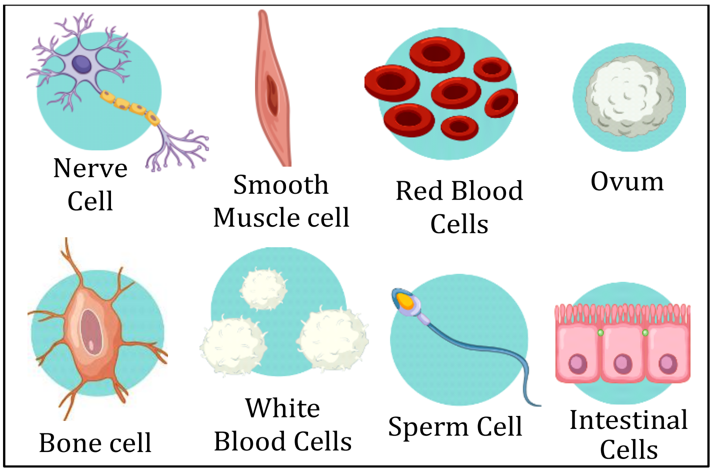
The size of organism is dependent upon the number of cells and not on the size of the cell.

SPOT LIGHT

Shape of Cell:

Shape of cell mainly depends upon the specific function it performs.

(i) Elongated & branched - Nerve cell
(ii) Discoidal/saucer - RBC
(iii) Spindle - Muscle cell
(iv) Spherical - Eggs
(v) Branched - Pigment cell of the skin
(vi) Slipper shaped - Paramecium
(vii) Cuboidal - Germ cells of gonads
(viii) Polygonal - Liver cells



Different shapes of cells of human body

Unicellular organisms vs multicellular organisms

The organisms which are made of single cell are called unicellular organisms. e.g. Bacteria, Amoeba, Paramecium, Chlamydomonas.

The single cell is able to perform all life processes, like obtaining of food, respiration, excretion, growth and reproduction.

The organisms which are made of number of cells are called multicellular organisms. e.g. some fungi, plants and animals.

All the cells of multicellular organisms have similar basic structure and undertake similar basic functions.

Non-living material in the cytoplasm of a cell such as pigment granules, starch granules, fat droplets, etc. are called cell inclusions.

SPOT LIGHT

➤ Generally, all cells are found to have the same organelles, irrespective of their function and location in an organism. A cell is able to live and perform all its functions because of these organelles.

Building Concepts 1

Which property of multicellular organisms differentiate them from unicellular organisms?

Explanation

Due to the property of division of labour the cell in multicellular organisms are specialised to perform different functions of the body and in unicellular organisms only a single cell has to perform all activities.

Check your Concepts 1

1. Who discovered cell and how?

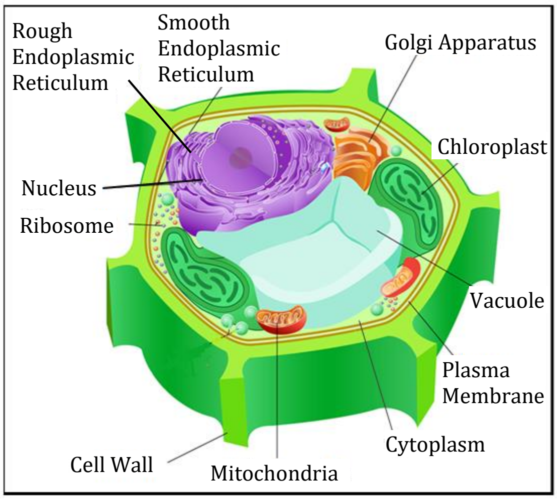
2. Why is the cell called structural and functional unit of life?

Basic structure of the cell

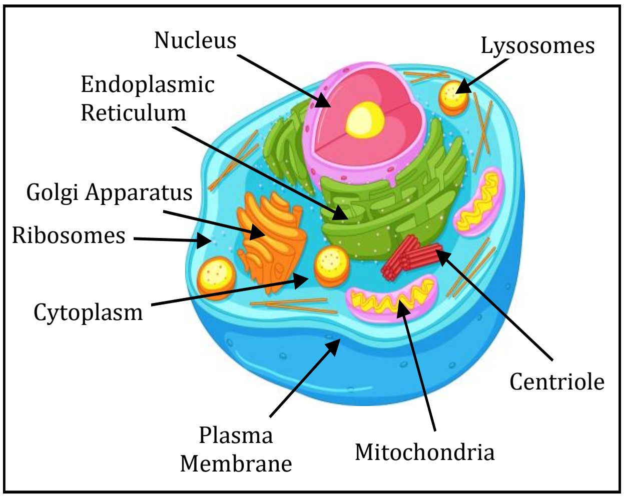
- Cell
  - Cell membrane
  - Protoplasm
    - Cytoplasm
      - Cytosol
      - Cell organelles
        - Single membranous
          - Endoplasmic Reticulum
          - Golgi body
          - Lysosomes
        - Double membranous
          - Plastid
          - Mitochondria
        - Non membranous
          - Ribosomes
          - Centrosomes
    - Nucleus
      - Nuclear membrane
      - Nucleoplasm
      - Nucleolus
      - Chromatin threads

Plant cells have a definite cell wall, contain plastids and a well-developed vacuole whereas in animal cells, cell wall and plastids are lacking but a centrosome is present.

SPOT LIGHT



Plant Cell



Animal Cell

Active Biology 1

Aim

To prepare stained temporary mount of onion peel cells.

Method

(i) Take an onion.

(ii) Separate out a thin onion scale from an onion. Tear it from the concave side to get a transparent, thin and membranous onion peel piece called epidermis.

(iii) Now keep this onion peel piece in a watch glass containing water.

(iv) Cut out a small portion of this peel and place it flat on a glass slide on a drop of water with the help of a thin camel-hair paint brush. Add a drop of safranin.

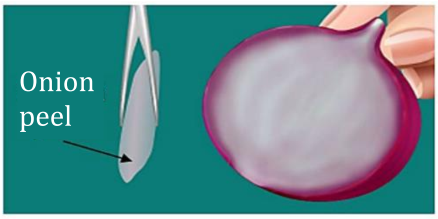
(v) Drain out the excess stain and mount the onion peel in a drop of glycerine under a coverslip.

(vi) Examine the slide under low and high powers of a compound microscope.

Observation

The epidermal cells of onion peel can be seen as regularly arranged linear or rectangular compartments with rigid cell walls. Nucleus is pushed towards the periphery due to presence of a central vacuole.


(i)

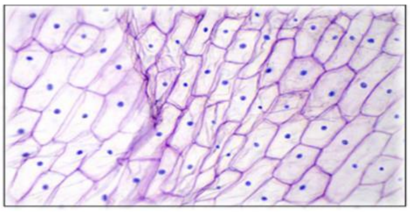


(ii)

(iii)

(iv)

(v)


Onion peel cells under microscope

Precautions

(i) Spread the peel uniformly on the slide.

(ii) Excess of stain should be drained off.

(iii) There should be no air bubble under the coverslip.

Be Alert !

Immediately put the peel of onion bulb in a water containing petri-dish to avoid its folding and drying.

Active Biology 2

Aim

To prepare a temporary stained slide of human cheek cells.

Method

(i) Wash your mouth and scrap a little of the internal lining of your mouth with a clean tooth pick or spatula or ice-cream spoon.

(ii) Place the scrapping in a watch glass containing a very small quantity of normal saline with the help of needle.

(iii) After cleaning, transfer the material to a glass slide.

(iv) Put a drop of methylene blue and wait for a couple of minutes. Wipe off the extra stain with a dropper or blotting paper.

(v) Put a drop of glycerine on the stained material.

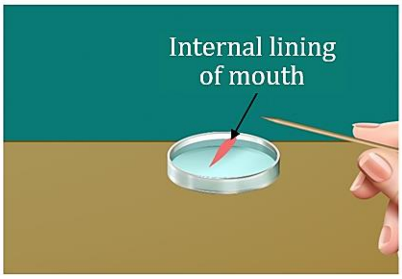
(vi) Place a coverslip. Tap the coverslip with the blunt end of needle so as to spread the cells.

(vii) Observe the temporary mount under low and high power of microscope.

Observation

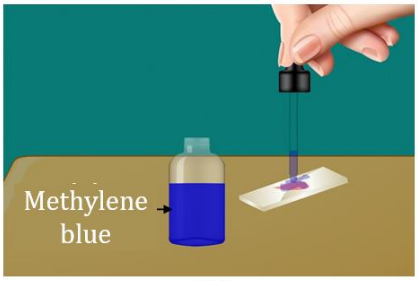
The cells are flat and polygonal in shape with distinct rounded nucleus in the middle. Each cell is bounded by a thin cell membrane. Cytoplasm is lightly stained.

(i)

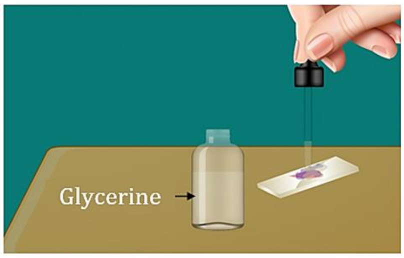


(ii)

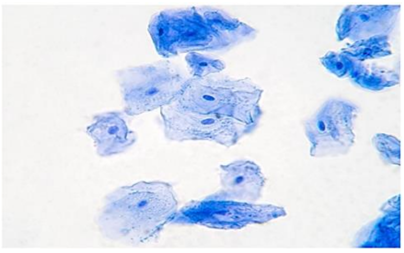
(iii)



(iv)



(v)



Stained slide of cheek cells of man

Precautions

(i) Scrapped material should be spread uniformly on the slide.

(ii) Excess of stain should be drained off.

(iii) There should be no air bubble under the coverslip.

Be Alert !

Do not scrap the cheek too hard as it may injure the buccal lining.

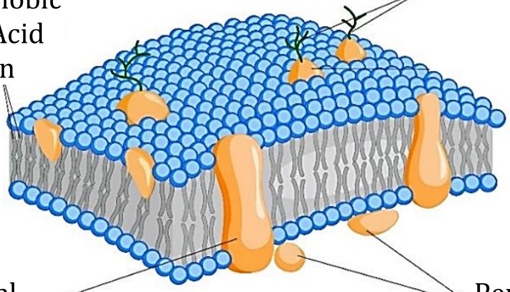
- These activities prove that not only cheek of human but all the organisms that you observe around are made up of cells. This confirms that the basic structural unit of life is cell.

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life

Plasma membrane or cell membrane or Plasmalemma

All living cells, have a plasma membrane that encloses their contents. It is flexible and made up of organic molecules called lipids, proteins and small fraction of carbohydrates. Flexibility enables the cell to engulf in food and other materials (endocytosis). e.g. Amoeba. It serves as selectively permeable barrier to the outside environment. The plasma membrane is permeable to specific molecules, however, it allows nutrients and other essential elements to enter the cell and waste materials to leave the cell. Small molecules such as oxygen, carbon dioxide are able to pass freely across the membrane by diffusion but the passage of larger molecules, such as amino acids and sugars, is carefully regulated.

> Plasma membrane is made of a bilayer of lipids (called phospholipids) and proteins. This structure was explained by Singer and Nicolson in the fluid mosaic model of plasma membrane.
>
> SPOT LIGHT



Plasma membrane

Plasma membrane performs the following functions:

(i) It protects the internal components of the cell.

(ii) It provides shape to the cell.

(iii) It allows materials to enter and leave the cell through the tiny holes called pores.

Note: Shrinking of protoplasm

In plant cell → plasmolysis

In animal cell → crenation

Building Concepts 2

What is the role of membrane carbohydrates in cell recognition?

Explanation

Short chains of sugars are linked to proteins and lipids on the exterior side of the plasma membrane, where they help in interaction with the surface molecules of other cells.

Check your Answers 1

1. Robert Hooke (1665) discovered cell. He used a primitive microscope to observe a cork slice. The scientist found that cork possesses a number of small box-like structures which he named cells (cellula which later abbreviated to cell). His work was published in the form of a book called Micrographia.

2. A living organism is made up of one or more cells. Therefore, cell is structural unit of life. All life functions of an organism reside in its cells. Cells may also become specialised to perform specific functions like contraction in muscle cell or impulse transmission in nerve cell. Therefore, cells are functional units of life also.

Transport across plasma membrane

In all living cells, substances may pass across the membrane with or without expenditure of energy. So, there are two types of transport:

(1) **Active transport:** Transport of substances across plasma membrane against the concentration gradient with expenditure of energy is called active transport.

(2) **Passive transport:** Transport of substances across plasma membrane along the concentration gradient i.e. from higher concentration to lower concentration without expenditure of energy is called passive transport.

Passive transport is of two types–

Passive transport
↓ Diffusion ↓ Osmosis

(i) **Diffusion**

The process of movement of substances (solid, liquid & gas) from the region of its higher concentration to the region of its lower concentration to spread uniformly in the given space is called diffusion.

Diffusion Across Cell Membrane

Metabolic gases (CO_2 and O_2) move out and into the cells through diffusion. Respiration in cell produces carbon dioxide. As the concentration of CO_2 (which is cellular waste) increases inside the cell as compared to the outside, CO_2 diffuses out of the cell into the external medium. Similarly, concentration of oxygen is always higher in the external medium as compared to the cell where it is being consumed in respiration. Therefore, oxygen diffuses from outside to the inside of cell.

(ii) **Osmosis**

Osmosis is the net diffusion of water across a selectively permeable membrane towards a higher solute concentration.

Example of osmosis:

(i) Absorption of water by plant roots.

(ii) Absorption of water by unicellular fresh water organisms.

Types of osmosis

Osmosis is of two types, endosmosis and exosmosis. Endosmosis is the osmotic entry of water into a cell or system. Exosmosis is the osmotic withdrawal of water from a cell or system.

Cell placed in solution

Plant and animal cells placed in salt or sugar solution will behave in one of the following ways depending upon the concentration of external solution.

> The process of taking in a bulk of materials from external environment into the cell is known as endocytosis. It is of two types-Phagocytosis (cell eating) and Pinocytosis (cell drinking).
> Phagocytosis + Pinocytosis → Endocytosis.
>
> **SPOT LIGHT**

(i) Hypotonic solution

The external solution is dilute as compared to cell contents. It has more water content while the water content is lower inside the cell. The cell membrane allows passage of water in both direction. Due to difference in concentration of water molecules, there is net flow of water molecules into the cells. The phenomenon is called endosmosis.

Building Concepts 3

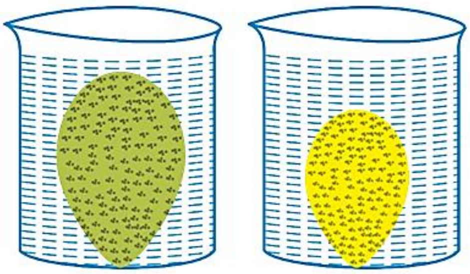
What will happen if both animal and plant cell are placed in hypotonic solutions?

Explanation

It causes swelling of animal cell, due to swelling, animal cell bursts. Plant cell do not burst due to presence of cell wall.

(ii) Isotonic solution

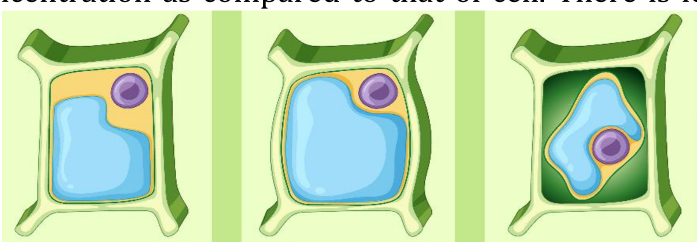
The external solution has the same concentration as that of the cell contents. Therefore, water content is equal on both sides. The amount of water going inside the cell is equal to the amount of water coming out of the cell. Therefore, there is no net movement of water. The cell size will remain the same.



Endosmosis Exosmosis

(iii) Hypertonic solution

The external solution has more concentration as compared to that of cell. There is less concentration of water in the external solution than inside the cell. Since more water is present inside the cell, more of it will pass out. The phenomenon is called exosmosis.



Normal cell placed in Isotonic solution | Normal cell placed in Hypotonic solution | Normal cell placed in Hypertonic solution

Plant Cell

Active Biology 3

Aim

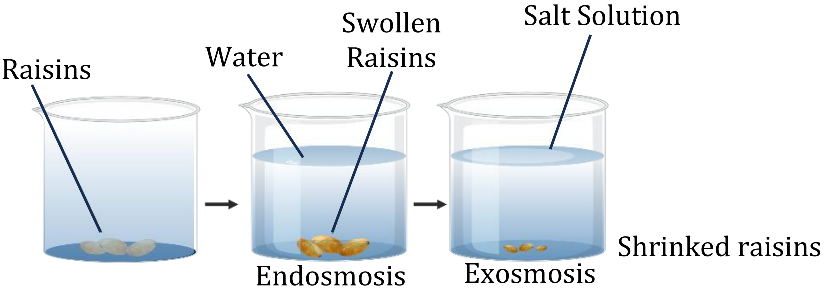
Demonstration of endosmosis and exosmosis.

Method

Put dry raisins or apricots in a petri dish having plain water. Observe after about 30 minutes.

Observation

Raisins or apricots swell up due to endosmosis. Some swollen raisins can be placed in concentrated sugar or salt solution. After some time, the raisins will shrink to previous form. It is due to exosmosis.



Dried raisins placed in water

Active Biology 4

Aim

Demonstration of endosmosis and exosmosis.

Method

Dip two equal sized raw eggs in dilute hydrochloric acid for an hour or so. The shells made up of calcium carbonate get dissolved. Wash the two eggs with water 2-3 times. Each egg is now surrounded by a membrane. Place one egg in water and the other in concentrated salt solution. Observe after 1 hour.

Observation

The egg placed in water swells up because of endosmosis. The egg placed in salt solution shrinks due to exosmosis.

Check your Concepts 2

1. How do substance like carbon dioxide and water move in and out of the cell?
2. Why plasma membrane is selectively permeable in nature?

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life

Building Concepts 4

What would happen if shelled raw egg and deshelled boiled egg are placed in water?

Explanation

There will be no change in size of the egg. In case of raw egg, the shell acts as an impermeable covering and the boiled egg does not show any change as its membranous covering has become dead.

Cell wall:

It is outermost rigid, freely permeable layer found outside the plasma membrane in all the plant cells, bacteria, blue-green algae, some protists and all fungi.

It is absent in animal cells. Cell wall permits the cells of plants, fungi and bacteria to withstand very dilute (hypotonic) external media without bursting. In such media the cells tend to take up water by osmosis. The cell swells, building up pressure against the cell wall. The wall exerts an equal pressure against the swollen cell. Because of their cell wall, such cells can withstand much greater changes in the surrounding medium than animal cells. When a living plant cell loses water through osmosis there is shrinkage or contraction of the contents of the cell away from the cell wall. This phenomenon is known as plasmolysis.

> Cellulose is a complex fibrous carbohydrate which cannot be digested by human beings and several other animals but can be digested by ruminant cattle due to presence of bacteria in their gut which secrete cellulase enzyme.
>
> **SPOT LIGHT**

Structure and function

Cell wall of plant cells is formed of a fibrous polysaccharide called cellulose, while it is formed of peptidoglycan in bacteria and blue-green algae. It is formed of chitin in most of fungi. Cell wall is a protective and supportive coat. It also provides a definite shape to the cell.

Building Concepts 5

Which of these is an essential component of cell, cell membrane or cell wall?

Explanation

A cell can exist without cell wall like animal cell but cannot exist without cell membrane because cell is a tiny mass of protoplasm covered by plasma membrane.

Active Biology 5

Aim

To prove the phenomenon of plasmolysis in the plant cells.

Method

(i) Prepare peels from the purple-coloured side of Rheo leaves by twisting and tearing.

(ii) Boil one peel for killing its cells. Keep the other peel fresh.

(iii) Mount the two peels on different slides in a drop of water.

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life

(iv) Observe the cells. They appear roughly similar on the two slides having cytoplasm and coloured sap.

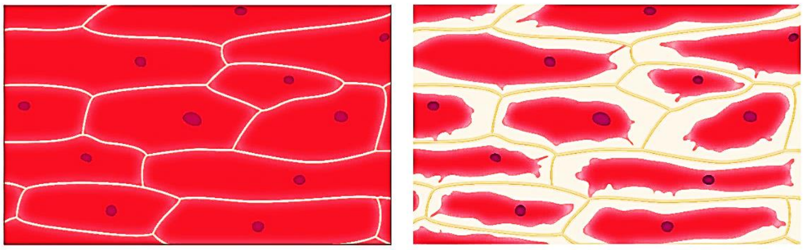
(v) Replace the drop of water with strong sugar or salt solution (hypertonic solution). For this a drop of solution is placed on one edge of cover slip and water is removed with the help of blotting paper from the edge.

(vi) Repeat once or twice to ensure that the peel dips in hypertonic solution.

(vii) Observe after 1-5 minutes.

Observation

When a plant cell or complete Rheo leaf is placed in a hypertonic solution, then exosmosis occurs from the central vacuole of the plant cell. The protoplasm starts separating from the cell wall and is completely separated. This process is called plasmolysis and the cell is called plasmolysed. This also proves the exosmosis. The green chloroplasts help one to see plasmolysis taking place more easily. But no plasmolysis occurs when Rheo leaf is first boiled in water for a few minutes and then placed in hypertonic solution. This shows that only living cells possess selectively permeable membranes and are therefore able to exhibit osmosis. Only living cells are able to absorb water by osmosis.

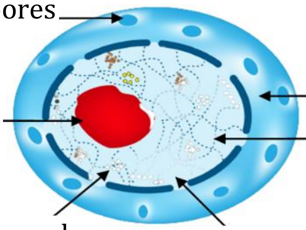


Plasmolysis

Nucleus (Headquarter of the cell)

Discovered by - Robert Brown (1831)

"Nucleus is double membrane bound dense protoplasmic body, which controls all cellular metabolism and encloses the genetic information of cell". Nucleus is considered as controller or director of the cell.



Structure of nucleus

Structure: It is made up of following four components

(a) Nuclear membrane/ Nuclear envelope/Karyotheca

(b) Nuclear matrix/Nucleoplasm

(c) Nucleolus

(d) Chromatin threads

Chromosomes are chemically made up of DNA and histone proteins. DNA carries all genetic information which is passed on to next generation. The functional segments of DNA are called genes.

SPOT LIGHT

(a) **Nuclear envelope:** Nucleus is surrounded by two membranes, that separates nucleoplasm from cytoplasm. The nuclear membrane has minute pores. These are called nucleopores. Nucleopore takes part in exchange of different substances between nucleoplasm and cytoplasm.

(b) **Nucleoplasm:** The part of protoplasm which is enclosed by nuclear membrane is called nucleoplasm. It contains chromatin threads and nucleolus.

(c) **Nucleolus:** Discovered by Fontana. Usually, one nucleolus is present in each nucleus but sometimes more than one nucleolus are present. It is a store house of RNA.

(d) **Chromatin threads:** A darkly stained network of long and fine threads called chromatin threads is present. Chromatin threads are intermingled with one another forming a network called chromatin reticulum. Whenever the cell is about to divide the chromatin material gets organized into chromosomes.

Functions of nucleus

The nucleus performs following functions:

(1) It controls all the metabolic activities of the cell.

(2) It brings about growth of the cell by directing the synthesis of structural proteins.

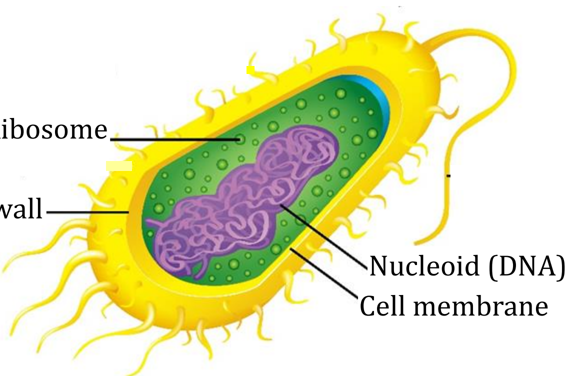
(3) It takes part in the formation of ribosomes.

(4) It regulates cell cycle.

(5) It contains genetic information and is concerned with the transmission of hereditary traits from one generation to another. On the basis of the presence or absence of a well-developed nucleus, organisms can be of two types–

> Nucleus is the largest cell structure. Human RBC lacks nucleus. Mature sieve tube in plants also lack nucleus.
>
> **SPOT LIGHT**

(1) **Prokaryotes:** Organisms whose cells lack a nuclear membrane and the genetic material (DNA, RNA) lies freely in the form of nucleoid, are known as prokaryotes. e.g. Bacteria, blue green algae.

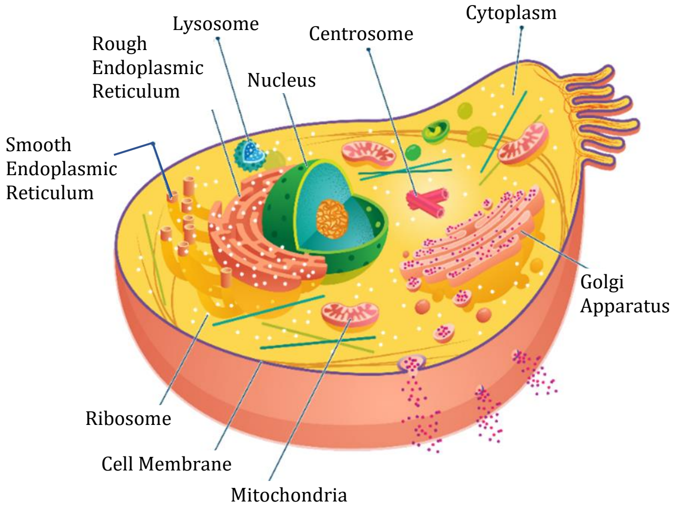


Prokaryotic Cell

(2) **Eukaryotes:** Organisms whose cells have a well-organized nucleus with nuclear membrane are known as eukaryotes. e.g. All plant and animal cells.

Note: Prokaryotic cells also lack most of the other cytoplasmic organelles present in eukaryotic cells. Many of the functions of such organelles are also performed by poorly organised parts of the cytoplasm. The chlorophyll in photosynthetic prokaryotic bacteria is associated with membranous vesicles (bag like structures) but not with plastids as in eukaryotic cells.

> The Nucleoid (meaning nucleus-like) is an irregularly shaped region within the cell of a prokaryote that contains all or most of the genetic material.
>
> **SPOT LIGHT**



Eukaryotic Cell

Cytoplasm

The fluid content inside the plasma membrane and excluding nucleus is called cytoplasm.

Cytoplasm has cytosol and many cell organelles.

Cytosol forms the aqueous, nearly transparent, structureless ground substance inside the cell.

Cell organelles are sub-cellular structures which have characteristic form, structure and function.

Almost all cell organelles are surrounded by membrane.

Building Concepts 6

The significance of membrane can be illustrated with the example of virus. How?

Explanation

Virus lack any membrane and hence do not show characteristics of life until they enter a living organism and use its cell machinery to multiply.

Cell Organelles

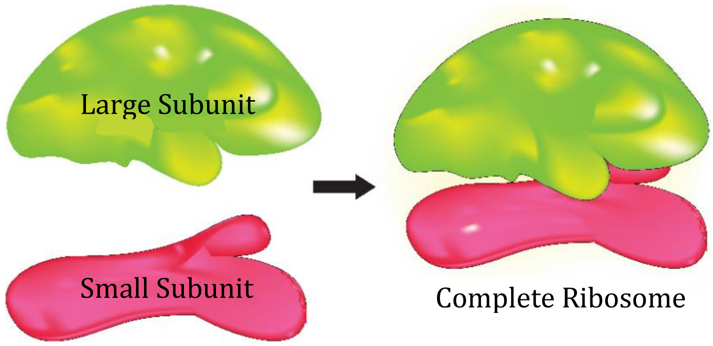
Ribosomes (Engine of cell / Protein construction teams)

Ribosomes are smallest and membrane less cell organelles. They are composed of RNA and proteins. It is also called organelle within an organelle and protein factory of cell.

Ribosomes are also called the protein builders or the protein synthesizers of the cell. These proteins might be used as enzymes or as support for other cell functions.

Ribosomes are found in many places around the cell. They are floating in the cytoplasm. Those floating ribosomes make proteins that will be used inside the cell. Other ribosomes are found on the endoplasmic reticulum (rough endoplasmic reticulum). These attached ribosomes make proteins that will be used inside the cell and for export out of the cell.

The 60S and 40S model works fine for eukaryotic cells. Prokaryotic cells have ribosomes made of 50S and 30S subunits.



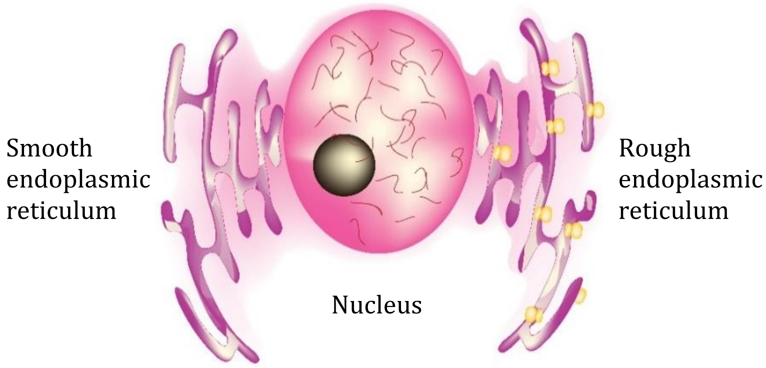
Ribosome

Endoplasmic Reticulum

Structure

Endoplasmic reticulum is a complex network of membrane bound channels or sheets, tubules and vesicles. It does not work alone. The ER works closely with the Golgi apparatus and ribosomes. It creates a network of membranes found through the whole cell. The ER may also look different from cell to cell, depending on the cell's function.

> The eukaryotic ribosomes are 80S while the prokaryotic ribosomes are 70S. Here 'S' (Svedberg's unit) stands for the sedimentation coefficient; it indirectly is a measure of density and size.
>
> **SPOT LIGHT**



Endoplasmic Reticulum

Components of E.R.

(i) **Cisternae:** These are long flattened and unbranched units arranged in stacks.

(ii) **Vesicles:** These are oval membrane bound structures.

(iii) **Tubules:** These are irregular, often branched tubes bounded by membrane. Tubules may be free or associated with cisternae.

> Steroid is a type of ringed organic molecules used for many purposes in an organism.
>
> **SPOT LIGHT**

Types of endoplasmic reticulum:

(i) **Smooth ER/Agranular ER:** It has smooth membranes which do not bear ribosomes. It acts as a storage organelle. It is important in the synthesis of lipids. They also store steroids. It is mainly made up of vesicles and tubules.

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life

(ii) **Rough ER/Granular ER:** It has rough membrane because a number of ribosomes are attached to its outer surface. RER is engaged in protein synthesis. It is mainly made up of cisternae. Tubules are very few.

Functions of endoplasmic reticulum

(i) ER serve as channels for the transport of materials (especially proteins) between various regions of cytoplasm or between cytoplasm and nucleus.

(ii) It also functions as a cytoplasmic framework providing a surface for some of the biochemical activities of the cell.

(iii) SER plays an important role in detoxifying many poisons and drugs and is also a site of lipid synthesis.

(iv) RER is engaged in synthesizing proteins.

(v) The proteins and lipids synthesized by RER and SER respectively help in building the cell membrane which is known as membrane biogenesis. Some of proteins and lipids formed by RER and SER functions as enzymes and hormones.

Check your Answers 2

1. Carbon dioxide moves into and out of the cell by diffusion while water does it through osmosis.

2. Cell membrane is selectively permeable membrane. It permits the entry of gases through diffusion. Ions, sugar, amino acids, etc. pass through the plasma membrane by an active process. Plasma membrane is impermeable to certain other materials. Therefore, it is selectively permeable.

Golgi complex (Traffic police of cell)

Discovered by Camillo Golgi (1898) in nerve cells of owl.

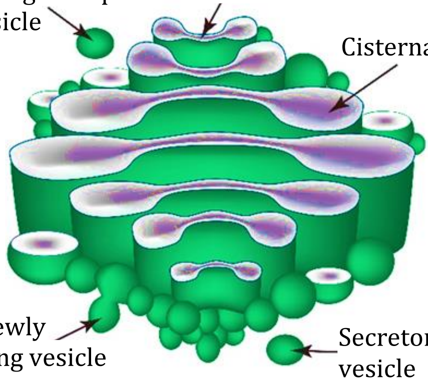
Golgi carried out a revolutionary method of staining individual nerve and cell structures. This method is referred to as "The black reaction". This method uses a weak solution of silver nitrate and is particularly valuable in tracing the processes and most delicate ramifications of cells. Golgi shared the Nobel prize in 1906 with Santiago Ramony Cajal for their work on structure of nervous system.

> Golgi apparatus is a cell organelle with single membrane covering. In plants, Golgi body is known as Dictyosomes, as they are found scattered in the plant cell.
>
> **SPOT LIGHT**

Position: It is located near the nucleus.

Golgi bodies are pleomorphic structures, because components of Golgi body are different in structure and shape in different cells.

➢ The Golgi body has a definite polarity. Concave side of Golgi body is always directed towards the cell membrane and is called maturation face or trans face while its convex surface is directed towards the nucleus and is called formative face or cis face. It appears that materials are transported from cis to trans face by vesicles.



Golgi complex

Structure: It is formed of four types of structural components.

(i) **Cisternae:** These are long flattened and unbranched saccules. 4 to 8 saccules are arranged in a stack. These are concentrically arranged near the nucleus.

(ii) **Tubules:** These are branched and irregular tube-like structures associated with cisternae.

(iii) **Vacuoles:** Large spherical structures associated to tubules.

(iv) **Vesicles:** Spherical structures arise by budding from tubules. Vesicles are filled with secretory materials.

Function

(i) It is involved in cell secretion and acts as storage, modification and condensation or packaging membrane.

(ii) It forms the acrosome of sperm. (Acrosome is a specialised vesicle present in the head of sperm filled with lytic enzymes which dissolve egg membrane at the time of fertilization.)

(iii) It also involved in the formation of lysosomes and secretory vesicles.

(iv) It is the site of formation of glycolipids and glycoproteins.

(v) Synthesis of cell wall material (polysaccharide synthesis).

(vi) Cell plate formation during plant cell division.

The cytoplasm surrounding Golgi body have fewer or no other organelles. It is called Golgi ground substance or zone of exclusion.

SPOT LIGHT

NOTE: Camillo Golgi was born at Corteno near Brescia in 1843. He studied medicine at the University of Pavia. After graduating in 1865, he continued to work in Pavia at the Hospital of St. Matteo. In 1872 he accepted the post of Chief Medical Officer at the Hospital for the Chronically Sick at Abbiategrasso. He first started his investigations into the nervous system in this hospital.

Building Concepts 7

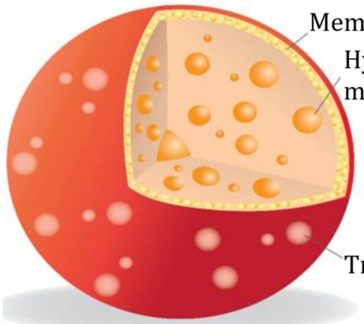
What would happen to the life of a cell if there was no Golgi apparatus?

Explanation

There would be no lysosomes for intracellular digestion and cleaning, no complexing of molecules, no excretion and no formation of plasma membrane.

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life

Lysosomes (Waste disposal system of the cell)



Lysosomes

Lysosomes are spherical bag like structures covered by single membrane. Lysosomes hold enzymes that were created by the cell. The purpose of the lysosome is to keep the cell clean by digesting any foreign materials as well as worn-out cell organelles. They might be used to digest food or break down the cell when it dies.

A lysosome is basically specialized vesicle that holds a variety of hydrolytic enzymes. The enzyme proteins are first created in the rough endoplasmic reticulum. Those proteins are packaged in a vesicle and sent to the Golgi apparatus. The Golgi then does its final work to create the digestive enzymes and pinches it off in the form of small specific vesicle. That vesicle is a lysosome. From there, the lysosomes float in the cytoplasm until they are needed.

- Lysosomes are called digestive bags because they contain lots of digestive enzymes.

Building Concepts 8

Why are lysosomes known as suicidal bags?

Explanation

At the time of cellular damage, the lysosomes burst to release its enzyme and they digest its own cell. Therefore, lysosomes are also known as suicidal bags of a cell.

> **SPOT LIGHT**
> Mitochondria are the sites where aerobic part of respiration called Kreb's cycle is performed.

Mitochondria (Powerhouse of cell)

Mitochondria are double membrane bound organelles of eukaryotic cells.

Occurrence

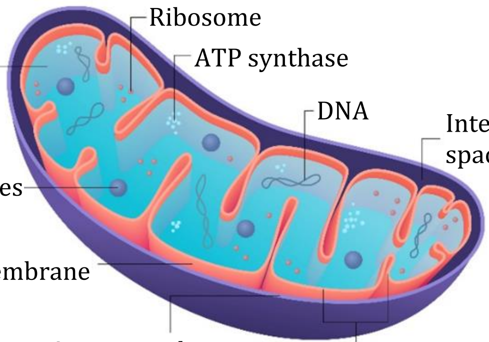
These are found in all eukaryotic cells except mature mammalian RBCs. These are absent in prokaryotic cells.

Shape

The shape may be fibrillar, spherical (in yeast), oval, sausage shaped, discoidal or cylindrical.

Origin

Mitochondria have a small and circular chromosome, few RNA and 70S ribosome of their own and make some of their own proteins, showing that they were once aerobic bacteria.



Mitochondria

Structure

Mitochondria are selectively autonomous organelles bound by an envelope of two units membranes and filled with a fluid matrix. The outer membrane is smooth and has porous proteins which forms channels for the passage of molecules through it. The inner membrane usually produces numerous infolds called cristae.

The cristae greatly increase the inner surface area of the mitochondria to hold a variety of enzymes. Cristae increases the surface area for ATP-generating chemical reaction. Cristae bear minute, regularly spaced tennis racket shaped particles known as F_1 particles or oxysomes. Oxysomes are concerned with ATP synthesis. Matrix contains various respiratory enzymes.

Function

Mitochondria are the main sites of cellular respiration. They bring about complete oxidation of food stuffs or respiratory substrates into carbon dioxide and water.

They are commonly known as 'power house of the cell' because they contain enzymes necessary for the complete oxidation of food and for release of high amount of energy in the form of ATP (Adenosine triphosphate) molecules. The body uses energy stored in ATP for synthesis of new chemical compounds and for mechanical work. ATP is also known as energy currency of the cell.

> Plastids and mitochondria are semiautonomous cell organelle as they are having their own DNA and ribosomes which are useful in protein synthesis.
>
> **SPOT LIGHT**

Plastids

Plastids are major organelles found in the cells of plants and algae. Plastids are the site of manufacture and storage of important chemical compounds used by the cell. Plastids often contain pigments used in photosynthesis and the types of pigments present can change or determine the cell's colour. Plastids are responsible for photosynthesis, storage of products like starch. They are double membranous cell organelles.

> Photoautotrophic bacteria possess photosynthetic pigments inside small vesicles which may be attached to plasma membrane.
>
> **SPOT LIGHT**

Origin: Like mitochondria, the chloroplast are self-duplicating organelles. They also have circular chromosome and 70S ribosome of their own. They also have a prokaryotic origin.

Plastids are of three types-

(i) Chromoplasts

It is responsible for pigment synthesis and storage. Chromoplasts are red, yellow and orange in colour and are found in petals of flower and in fruits.

Their colour is due to two pigments, carotene and xanthophyll.

(ii) Leucoplasts

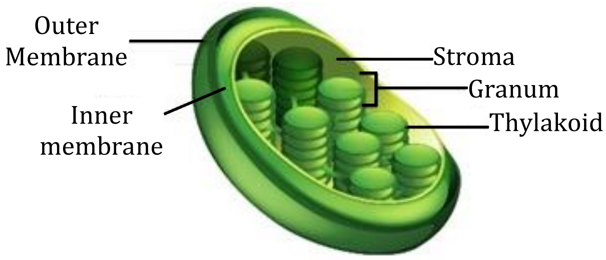
Leucoplasts are colourless or white plastid. They occur in plant cell not exposed to light, such as roots and seeds. Leucoplasts are the centre of starch grain formation and they are also involved in the storge of oil, starch and proteins.

(iii) Chloroplasts

Chloroplasts are probably the most important among the plastids since they are directly involved in photosynthesis. They are usually situated near the surface of the cell and occur in those parts that receive sufficient light. e.g. the palisade cells of leaves. The green colour of chloroplasts is caused by the green pigment chlorophyll. It also contains various yellow-orange pigment in addition to chlorophyll.

Structure of chloroplast

Chloroplasts are usually disc-shaped and surrounded by a double membrane. Inside the inner membrane, there is a watery protein-rich ground substance called as stroma, in which, there is embedded a continuous membrane system, the granal network. This network forms a three-dimensional arrangement of membrane bound vesicles called thylakoids. The thylakoids usually lie in stacks called grana and contain the photosynthetic pigments - green chlorophyll a and b and the yellow to red carotenoids. The grana are interconnected by tubular membranes called lamellae.



Chloroplast

Function of Chloroplast

Chloroplasts are the sites of photosynthesis. They contain enzymes and coenzymes necessary for the process of photosynthesis.

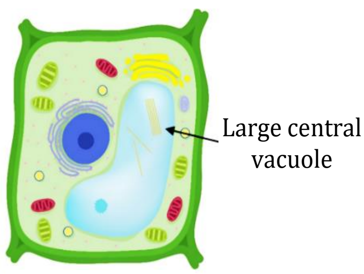
Check your Concepts 3

1. Name the cell organelle in which following structures are present.
   (A) Cristae (B) Stroma (C) Centriole (D) Chromosome
2. Name the type of plastid which stores food.
3. If the cellular organization is destroyed due to some physical or chemical influence. What will happen?
4. Where are proteins synthesized inside the cell?

Vacuoles:

Vacuoles are membrane bound storage sacs that contain non-living liquid or solid contents.

In animal and young plant cells, vacuoles are small. In mature plant cells, there is a large central vacuole occupying 50–90% of cell volume and contain cell sap. The covering membrane of the vacuole is called tonoplast.



Plant Cell

Functions

(i) In plant cell, vacuoles are full of cell sap and provide turgidity and rigidity to the cell.

(ii) Vacuoles store amino acid, sugars, various organic acids and some proteins.

(iii) They act as dump house for excretory products in plant cells.

Note: In single-celled organisms like Amoeba, the food vacuole contains the food items that the Amoeba has consumed. In some unicellular organisms, specialised vacuoles also play important roles in expelling excess water and some wastes from the cell.

Contractile vacuole in unicellular fresh water organisms take part in osmoregulation and excretion.

SPOT LIGHT

Comparison between Animal cell and Plant cell

	Animal Cell	Plant Cell
Cell wall	Absent	Present
Plasma membrane	Present	Present
Lysosomes	Lysosomes occur in cytoplasm.	Lysosomes usually not evident.
Nucleus	Present in center	Present in periphery
Shape of the cell	Round (irregular shape)	Rectangular (fixed shape)
Chloroplast	Animal cells don't have chloroplast.	Plant cells have chloroplasts because they make their own food.
Cytoplasm	Present	Present
Endoplasmic Reticulum (Smooth and Rough)	Present	Present
Ribosomes	Present	Present
Mitochondria	Present	Present
Vacuole	One or more small vacuoles (much smaller than plant cells).	One, large central vacuole can occupy upto 90% of cell volume.
Centrioles	Present in all animal cells.	Only present in lower plant forms.
Golgi Apparatus	Present	Present and are called dictyosomes.

Check your Answers 3

1. (A) Mitochondria (B) Chloroplast
(C) Centrosome (D) Nucleus

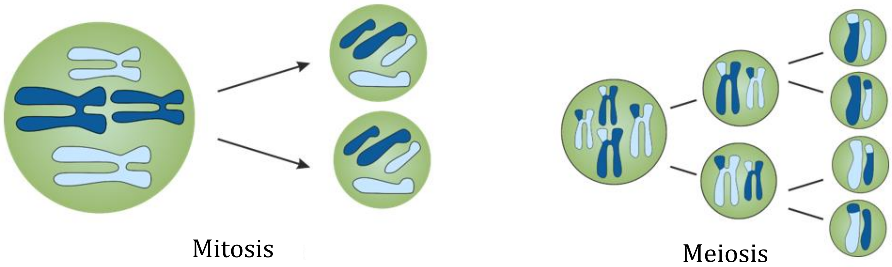
2. Leucoplast

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life

3. Lysosomes will burst to release digestive enzymes. Digestive enzymes cause break down of various cellular component causing destruction of the cell.

4. Ribosome present in the cytoplasm and Rough endoplasmic reticulum synthesise proteins.

Cell division:



Mitosis Meiosis

New cells are formed in organisms in order to grow, to replace old, dead and injured cells, and to form gametes required for reproduction. The process by which new cells are made is called cell division. There are two main types of cell division: mitosis and meiosis.

The process of cell division by which most of the cells divide for growth is called mitosis. In this process, each cell called mother cell divides to form two identical daughter cells. The daughter cells have the same number of chromosomes as mother cell. It helps in growth and repair of tissues in organisms.

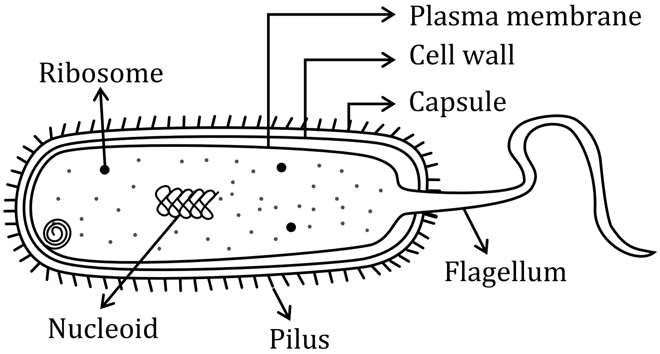
Specific cells of reproductive organs or tissues in animals and plants divide to form gametes, which after fertilisation give rise to offspring. They divide by a different process called meiosis which involves two consecutive divisions. When a cell divides by meiosis it produces four new cells instead of just two. The new cells only have half the number of chromosomes than that of the mother cells.

Note: Centrosome: Centrioles and centrosphere are collectively called centrosome. Two centrioles are located just outside the nucleus and lie at right angle (90°). It initiates cell division by arranging spindle fibers between 2 poles of cell.

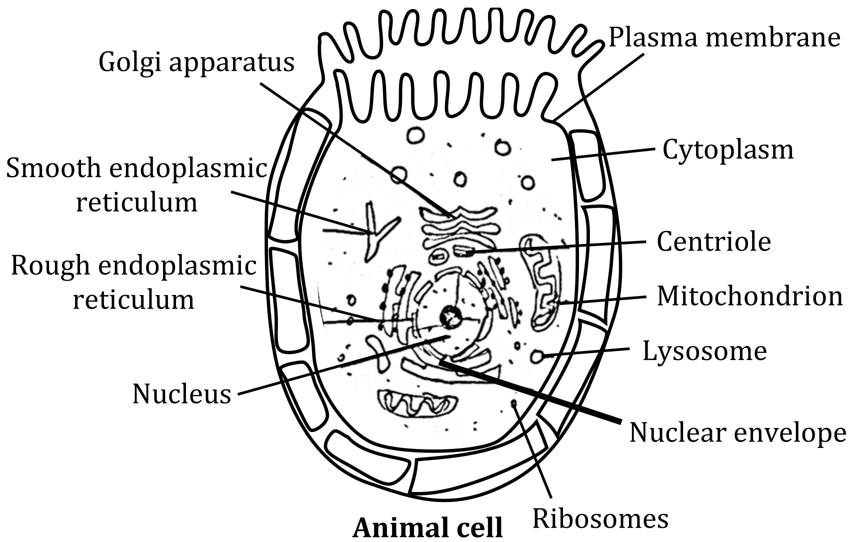
Comparison between Prokaryotes & Eukaryotes

Characteristic	Prokaryotes	Eukaryotes
Size of cell	Typically 1-10 μm in diameter	Typical 5-100 μm in diameter
Nucleus	No nuclear membrane or nucleoli	True nucleus, consisting of nuclear membrane and nucleoli
Membrane enclosed cell organelles	Absent	Present; examples include Lysosomes, Golgi complex, Endoplasmic reticulum Mitochondria, Chloroplasts
Cell Wall	Usually present ; When present, made up of peptidoglycan	When present, made up of cellulose or chitin
Ribosomes	70S type of ribosome	80S type of ribosome
DNA	Single circular DNA	Linear DNA organised into multiple chromosomes

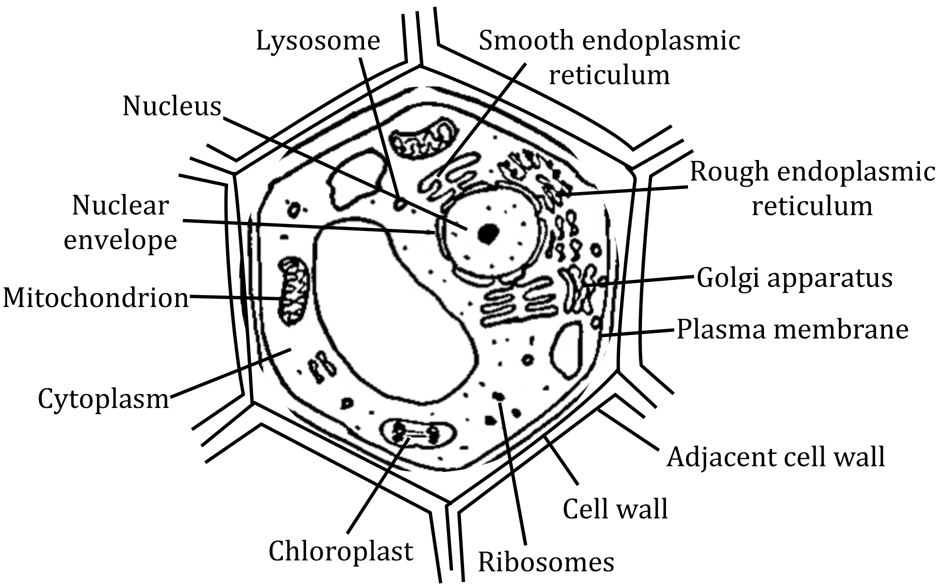
Biology Diagrams made Easy



Prokaryotic cell

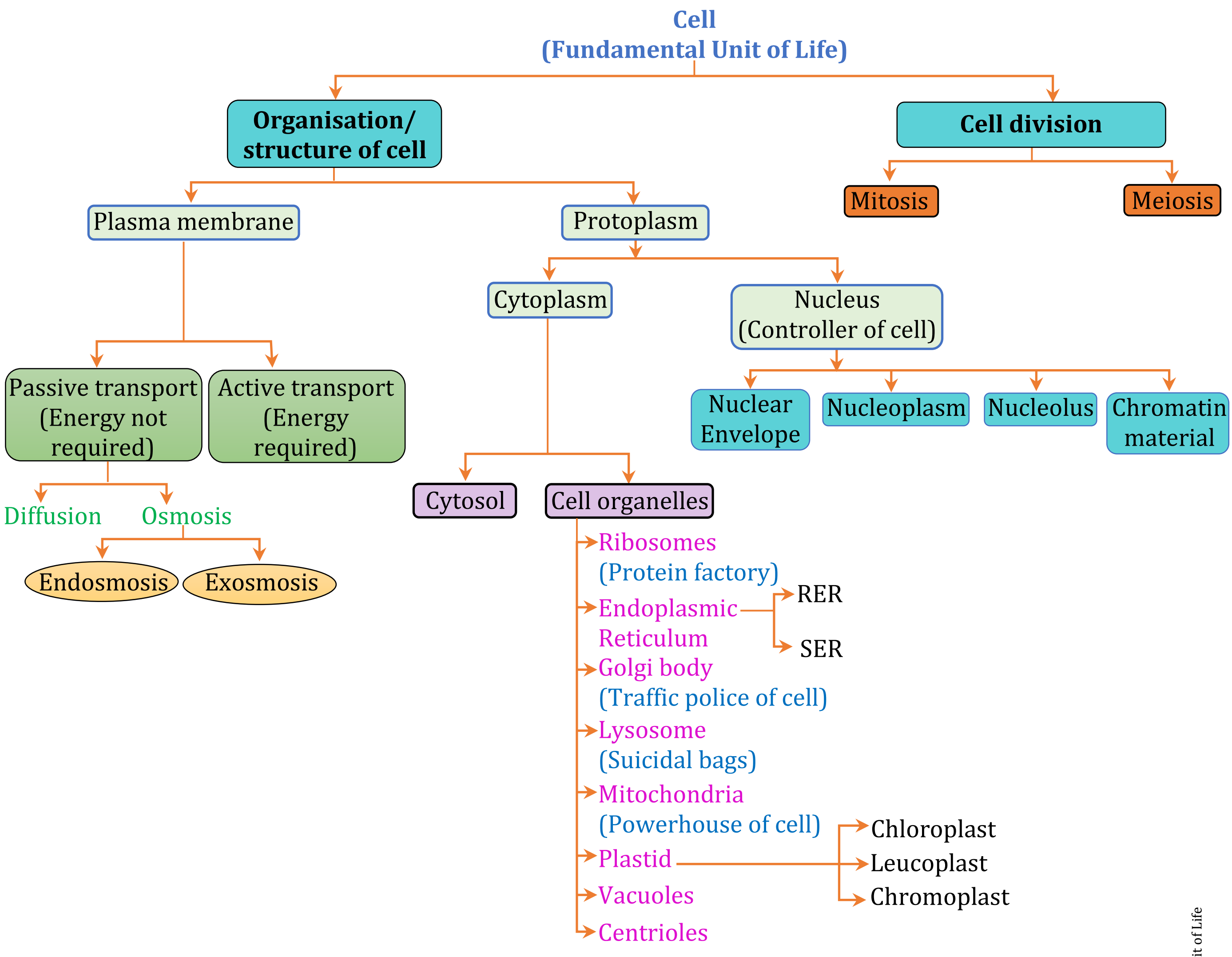


Animal cell



Plant cell

Chapter At a Glance



SOME BASIC TERMS

1. **Magnifying lens :** Glass used to produce a bigger image of an object.
2. **Objective lens :** Lens nearest to object.
3. **Eye piece :** Lens nearest to eye.
4. **Petri dish :** A covered dish that is not very deep.
5. **Periphery :** The outer edge of particular area.
6. **Synthesized :** Prepared
7. **Stack :** A pile of something.
8. **Turgid :** Swollen and distended.
9. **Permeable :** Allowing a liquid or gas to pass.
10. **Biogenesis :** The synthesis of chemical compounds or structures in the living organism.
11. **Prominent :** Noticeable ; easy to see.
12. **Longitudinal :** Running lengthwise.
13. **Mesophyll :** Chlorophyll containing cells.
14. **Tendril :** A long thin part that grows from a climbing plant for support.
15. **Leaf stalk :** A thin long structure which attaches leaves to a stem.
16. **Epidermis :** The surface epithelium of skin/outer layer of tissue in plant.
17. **Protoplasm :** The fluid content of the cell inner to plasma membrane and including nucleus is called cytoplasm.
18. **Nucleoid :** Irregularly shaped region in prokaryotic cell where genetic material is present.
19. **Tonoplast :** Covering membrane of vacuole.
20. **Vesicle :** Membraneous structure that stores cellular product.

Publishing$\PNCF\2024-25\Print Module\SET-1\NCERT\Biology\9th\01. The Fundamental Unit of Life